LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

WHY THIS MATTERS Miguel is interested in joining a recreation center. Recreation World charges the monthly fee shown on the flyer. How can Miguel determine the cost of being a member of Recreation World for any number of months?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
equation *Spanish: ecuación*	A mathematical sentence with an equals sign that shows two amounts are the same.	y = 5x
variable *Spanish: variable*	A letter that represents a quantity that can change, or vary.	x, y, c, h
rate *Spanish: tasa*	A fixed amount of one quantity for every one unit of the other quantity.	$3 for each pound
represent *Spanish: representar*	To show a relationship using a tool such as a table, a graph, or an equation.	The same membership shown as a table, a graph, AND c = 24.95m.
define the variables *Spanish: definir las variables*	To state exactly what quantity each letter in an equation stands for.	c = total cost
Constant *Spanish: Constante*	A number in an equation that never changes.	the 5 in y = 5x

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Write an equation for the pairs (1, 6), (2, 12), (5, 30).

1. 6 ÷ 1 = 6, 12 ÷ 2 = 6, 30 ÷ 5 = 6.
2. The rate is 6 every time.
3. Check: 6 × 5 = 30.

Answer: y = 6x

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A canoe rents for $15 per hour. Which quantity is independent — hours or total cost?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In my equation, x stands for _____ and y stands for _____ .
- The constant rate is _____ because every y value is _____ .
- My equation is y = _____ , which means _____ .
- I checked my equation by substituting _____ for x.

Word bank equation · variable · rate · constant · define · substitute · times · per · table · row · relate · check

WATCH OUT!

A common mistake

Writing the equation with addition

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Miguel is interested in joining a recreation center. Recreation World charges the monthly fee shown on the flyer. How can Miguel determine the cost of being a member of Recreation World for any number of months?

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

In the equation c = 24.95m for the gym membership, what does m represent?

(A) The number of months of membership
(B) The total cost in dollars
(C) The monthly fee of $24.95
(D) The name of the gym

HINT Which letters are variables here — and which quantity does each stand for?

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Recreation World charges $24.95 per month. What is the total cost of 2 months of membership?

(A) $24.95
(B) $26.95
(C) $49.90
(D) $74.85

HINT What does ONE month cost?

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

A canoe rents for $15 per hour. Let h = hours and c = total cost. Which equation represents the relationship?

(A) c = 15h
(B) c = 15 + h
(C) h = 15c
(D) c = h − 15

HINT Say it in words: the total cost equals $15 ___ the number of hours.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Using c = 15h, what is the total cost of a 3-hour canoe rental?

(A) $15

(B) $18

(C) $45

(D) $75

HINT Replace h with 3 in the equation.

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

5 **Circle the letter of the best answer. Show how you know.**

In the gym membership situation, which quantity is the DEPENDENT variable?

(A) The number of months

(B) The total cost

(C) The monthly fee of $24.95

(D) The number of gyms in town

HINT Which quantity responds to the other?

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

The membership table shows: 1 month, $24.95; 2 months, $49.90; 3 months, $74.85. By how much does the total cost increase each month?

(A) $24.05

(B) $24.95

(C) $25.00

(D) $49.90

HINT Subtract one row's total from the next row's total.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both equations charge by a rate. What do m and h each stand for in their own situation, and why would c = 24.95m be wrong for the canoe?

SENTENCE STARTER In ___, ___ stands for ___, so the rate ___ means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Miguel is interested in joining a recreation center. Recreation World charges the monthly fee shown on the flyer. How can Miguel determine the cost of being a member of Recreation World for any number of months?

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A table pairs months with total cost: (1, 24.95), (2, 49.90), (4, 99.80). Which equation matches the table, with m = months and c = cost?

(A) c = 24.95m
(B) c = m + 24.95
(C) m = 24.95 + c
(D) c = 49.90m

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Using c = 24.95m, what is the total cost of 10 months of membership?

(A) $34.95
(B) $224.55
(C) $249.50
(D) $2,495.00

My work — one step per line	What I did to both sides

3 Circle the letter of the best answer. Show how you know.

Using c = 15h, what is the total cost of a 5-hour canoe rental?

(A) $20
(B) $60
(C) $75
(D) $80

My work — one step per line	What I did to both sides

4 Circle the letter of the best answer. Show how you know.

On the canoe graph with hours on the horizontal axis and cost on the vertical axis, which point lies on the graph?

(A) (4, 60)
(B) (4, 19)
(C) (4, 55)
(D) (60, 4)

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

The friends have $90 to spend. Using c = 15h, for how many hours can they rent the canoe?

(A) 5 hours
(B) 6 hours
(C) 75 hours
(D) 1,350 hours

My work — one step per line	What I did to both sides

6 **Circle the letter of the best answer. Show how you know.**

How are the canoe graph and the equation c = 15h related?

(A) The graph and the equation show different relationships

(B) The equation only works for the points shown on the graph

(C) Every point on the graph is a pair (h, c) that makes the equation true

(D) The graph replaces the equation, so only one can be used

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both equations charge by a rate. What do m and h each stand for in their own situation, and why would c = 24.95m be wrong for the canoe?

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LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Challenge · Extension

MASTERY CHECK
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- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each scenario to the equation that models it.

_____ Canoe rental: $15 per hour

_____ Gym membership: $24.95 per month

_____ Football tickets: $45 each

_____ Tram climbing 1,200 feet per minute

(A) c = 15h

(B) c = 24.95m

(C) c = 45t

(D) d = 1200m

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Miguel wants the cost of a full year (12 months) at Recreation World. Using c = 24.95m, what is the total?

(A) $36.95

(B) $249.50

(C) $299.40

(D) $300.00

My work — one step per line	What I did to both sides

3 **Circle the letter of the best answer. Show how you know.**

A different gym charges $19.95 per month. How does Miguel's equation change for this gym?

(A) The variable changes: c = 24.95x instead of c = 24.95m

(B) The rate changes: c = 19.95m instead of c = 24.95m

(C) The operation changes: c = 19.95 + m

(D) Nothing changes — all gyms have the same equation

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the benefit of an equation over a table for the membership costs?

(A) The equation lists more rows than the table

(B) The equation is the only representation that shows the cost increasing

(C) The table is always wrong and the equation is always right

(D) The equation gives the total cost for ANY number of months, not just the rows listed

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The friends spent exactly $120 on the canoe. Solve 15h = 120. How long was the rental?

(A) 8 hours

(B) 9 hours

(C) 105 hours

(D) 1,800 hours

My work — one step per line	What I did to both sides

EXPLAIN YOUR THINKING

Both equations charge by a rate. What do m and h each stand for in their own situation, and why would c = 24.95m be wrong for the canoe?

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